# MORPHISM PROPERTIES ARE THICKENING-INVARIANT (13-CASE TRANSFER)

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 576-597 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 599-649, the complete author proof (49 body lines).

*Editorial note.* Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 18 of queue_stacks-stacks-more-morphisms.json (pb=2468, stmt 576-597, proof 599-649).

*Editorial note.* stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 09ZU: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 364–373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-thickening: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 308–316. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-thickening-diagonals: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 454–468. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-thickening-properties: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 507–510. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 09ZY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 649–649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04YW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 649–649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0AHJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 649–649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04XD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 649–649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $(f, f') : (\mathcal{X} \subset \mathcal{X}') \to (\mathcal{Y} \subset \mathcal{Y}')$ be a morphism of thickenings of algebraic stacks. Then*

1. *$f$ is an affine morphism if and only if $f'$ is an affine morphism,*
2. *$f$ is a surjective morphism if and only if $f'$ is a surjective morphism,*
3. *$f$ is quasi-compact if and only if $f'$ quasi-compact,*
4. *$f$ is universally closed if and only if $f'$ is universally closed,*
5. *$f$ is integral if and only if $f'$ is integral,*
6. *$f$ is universally injective if and only if $f'$ is universally injective,*
7. *$f$ is universally open if and only if $f'$ is universally open,*
8. *$f$ is quasi-DM if and only if $f'$ is quasi-DM,*
9. *$f$ is DM if and only if $f'$ is DM,*
10. *$f$ is (quasi-)separated if and only if $f'$ is (quasi-)separated,*
11. *$f$ is representable if and only if $f'$ is representable,*
12. *$f$ is representable by algebraic spaces if and only if $f'$ is representable by algebraic spaces,*
13. *add more here.*

*Proof.* By Lemma lemma-thickening the morphisms $\mathcal{X} \to \mathcal{X}'$ and $\mathcal{Y} \to \mathcal{Y}'$ are universal homeomorphisms. Thus any condition on $|f| : |\mathcal{X}| \to |\mathcal{Y}|$ is equivalent with the corresponding condition on $|f'| : |\mathcal{X}'| \to |\mathcal{Y}'|$ and the same is true after arbitrary base change by a morphism $\mathcal{Z}' \to \mathcal{Y}'$. This proves that (2), (3), (4), (6), (7) hold.

In cases (8), (9), (10), (12) we can translate the conditions on $f$ and $f'$ into conditions on the diagonals $\Delta$ and $\Delta'$ as in Lemma lemma-thickening-diagonals. See Morphisms of Stacks, Definition 04YW and Lemma 0AHJ. Hence these cases follow from Lemma lemma-thickening-diagonals.

Proof of (11). If $f'$ is representable, then so is $f$, because for a scheme $T$ and a morphism $T \to \mathcal{Y}$ we have $\mathcal{X} \times_{\mathcal{Y}} T = \mathcal{X} \times_{\mathcal{X}'} (\mathcal{X}' \times_{\mathcal{Y}'} T)$ and $\mathcal{X} \to \mathcal{X}'$ is a closed immersion (hence representable). Conversely, assume $f$ is representable, and let $T' \to \mathcal{Y}'$ be a morphism where $T'$ is a scheme. Then

$$\mathcal{X} \times_{\mathcal{Y}} (\mathcal{Y} \times_{\mathcal{Y}'} T') = \mathcal{X} \times_{\mathcal{X}'} (\mathcal{X}' \times_{\mathcal{Y}'} T') \to \mathcal{X}' \times_{\mathcal{Y}'} T'$$

is a thickening (by Lemma 09ZU) and the source is a scheme. Hence the target is a scheme by Lemma lemma-thickening-properties.

In cases (1) and (5) if either $f$ or $f'$ has the stated property, then both $f$ and $f'$ are representable by (11). In this case choose an algebraic space $V'$ and a surjective smooth morphism $V' \to \mathcal{Y}'$. Set $V = \mathcal{Y} \times_{\mathcal{Y}'} V'$, $U' = \mathcal{X}' \times_{\mathcal{Y}'} V'$, and $U = \mathcal{X} \times_{\mathcal{Y}'} V'$. Then the desired results follow from the corresponding results for the morphism $(U \subset U') \to (V \subset V')$ of thickenings of algebraic spaces via the principle of Properties of Stacks, Lemma 04XD. See More on Morphisms of Spaces, Lemma 09ZY for the corresponding results in the case of algebraic spaces. □